

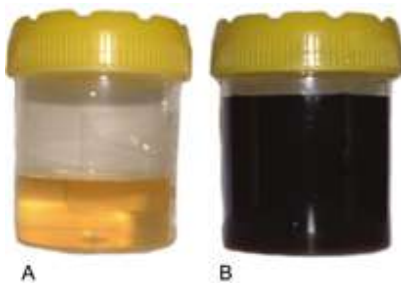
ALBINISM

- ◆ Tyrosinase catalyzes the conversion of Tyrosine $\rightarrow$ DOPA $\rightarrow$ Melanin
- ◆ In albinism, a deficiency or absence of tyrosinase results in no melanin synthesis

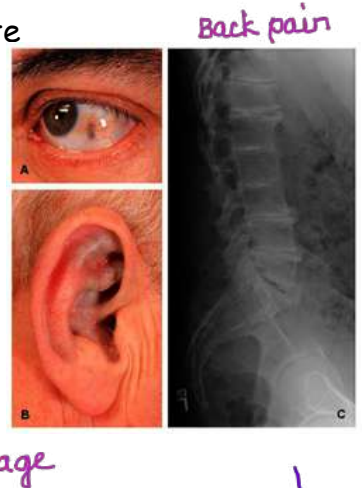
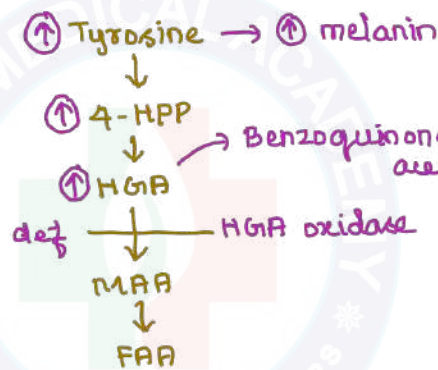


ALKAPTONURIA ~30-40yr | Backpain | pigmentation

- ◆ Homogentisate oxidase converts homogentisic acid $\rightarrow$ maleylacetoacetate
- ◆ Deficiency leads to accumulation of homogentisic acid.



Rx:- Nitisidone



Pellagra

- ◆ Tryptophan $\rightarrow$ (converted into) $\rightarrow$ Niacin (Vitamin B3)
- > This conversion requires Vitamin B6 (Pyridoxine) as a cofactor

Cause of Pellagra:

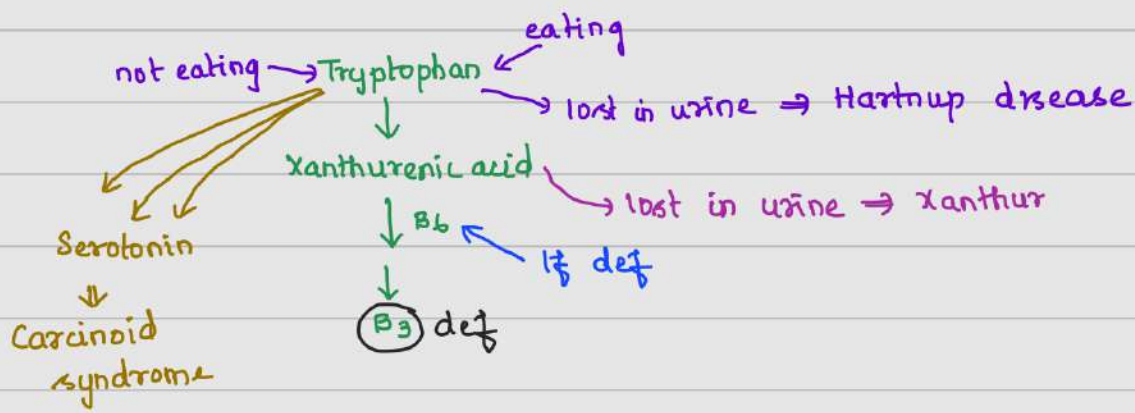
Deficiency of Niacin (Vitamin B3)

- ◆ Dietary deficiency (in maize/corn-based diets that lack tryptophan and niacin)
- ◆ Hartnup disease (↓tryptophan reabsorption in renal tubules)
- ◆ Carcinoid syndrome (↑tryptophan shunted to serotonin synthesis)
- ◆ Vitamin B6 deficiency (impaired conversion of tryptophan to niacin)

The "3 Ds" of Pellagra:

1. Dermatitis - Photosensitive, pigmented rash (sun-exposed areas; Casal necklace around the neck)
2. Diarrhea
3. Dementia
4. If untreated, it progresses to the 4th D: Death





PKU

neonate + bad smell in urine → MR

alt def of $\left\{ \begin{array}{l} \text{PA-Hydroxylase} \leftarrow \text{PKU-1} \\ \text{BH}_2 \text{ reductase} \leftarrow \text{PKU-2} \end{array} \right.$

IOC:- Tandem mass spectrometry.

Urine → sweet smell + pancake syrup smell

→ 4. maple syrup urine disease

def of branch chain

α -ketoacid dehydrogenase

mainly its defect of oxi-decarboxylation

Urine $\left\{ \begin{array}{l} \rightarrow \text{sweaty feet} \Rightarrow \text{Isovaleric acidemia} \\ \rightarrow \text{Boiled cabbage} \Rightarrow \text{Tyrosinemia (I)} \\ \rightarrow \text{Swimming pool water} \Rightarrow \text{Hawkinsinuria} \end{array} \right.$

Dopamine / NE / E
melanine
T₃ / T₄

} Tyrosine

Serotonin / B₃ /
melatonin

} Tryptophan

↳ Glutathione → Glycine / cystine / Glutamate.

↳ Histidine $\xrightarrow{-\text{CO}_2}$ Histamine

↳ Arginine → Nitric oxide

↳ Glycine
Arginine
Methionine } → Creatinine.

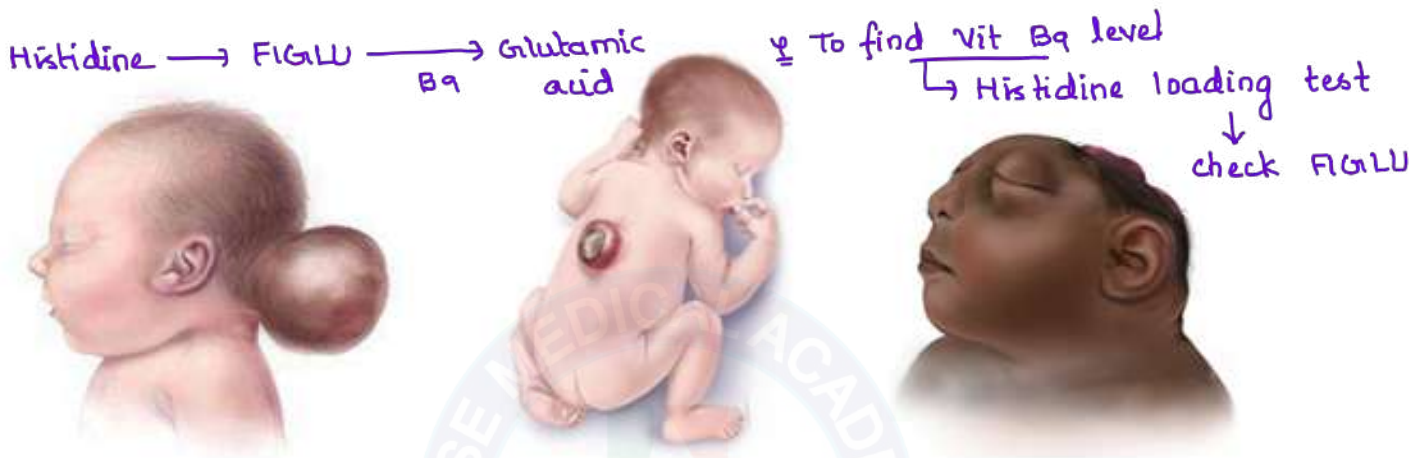
Hartnup Disease:

- ◆ Defect in neutral amino acid tryptophan transport in kidney and intestine
- ◆ Leads to loss of tryptophan in urine --> ↓ Niacin synthesis — Pellagra
- ◆ Maize (corn) eaters: Corn is low in niacin and tryptophan

Neural Tube Defects (NTDs)

Neural Tube Defects (NTDs)

Folate is essential for DNA synthesis and cell division, especially during early embryonic development. Deficiency impairs neural tube closure.



Most Common Type:

- ◆ Spina bifida (specifically spina bifida occulta is the mildest; meningocele is more severe)
- ◆ Multifactorial inheritance

Combination of genetic predisposition and environmental factors

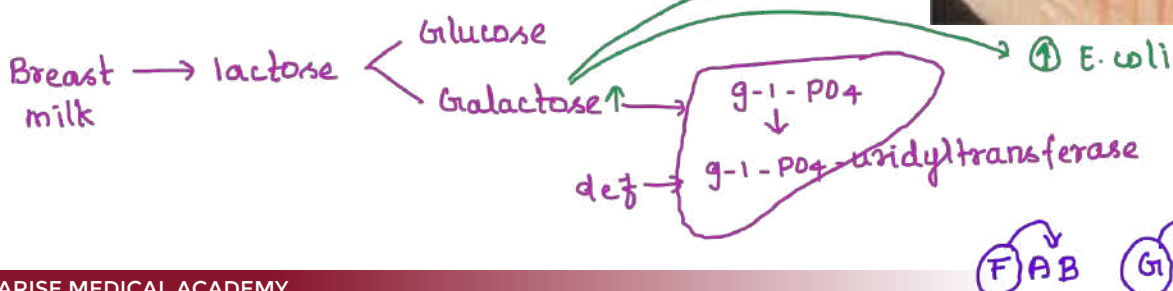
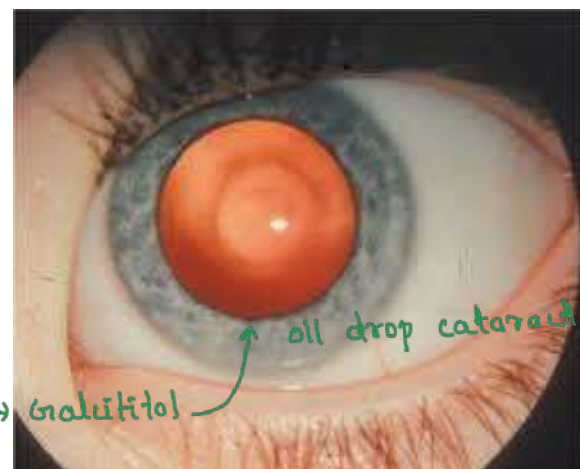
Galactosemia [B-feed → flb → vomiting + jaundice]

Due to Accumulation of: Galactitol (also known as dulcitol)

When enzymes like galactose-1-phosphate uridylyltransferase (GALT) are deficient (as in classic galactosemia), galactose and galactitol accumulate.

B/L Oil Drop Cataract:

- ◆ Caused by accumulation of galactitol in the lens
 - ◆ Galactitol is osmotically active, drawing water into the lens --> lens fibers swell --> opacity.
- It is only cataract which is reversible.



☞ Fruits → F/B → vomiting, jaundice + Hypoglycemia
 Δ:- Fructose intolerance → def ⇒ Aldolase (B)

Phrynoderma

- ◆ A form of follicular hyperkeratosis presenting as hyperkeratotic follicular papules on extensor surfaces (elbows, knees, buttocks).
- ◆ Commonly referred to as "toad skin" due to its rough, bumpy appearance.
- ◆ Most Common Vitamin Deficiency (especially in India): Vitamin A deficiency (Retinol)



Acrodermatitis Enteropathica : → Zinc deficiency.

- ◆ Acrodermatitis enteropathica is a rare autosomal recessive disorder caused by impaired zinc absorption in the intestine.
- ◆ SLC39A4 Gene.
- ◆ Affects periorificial and acral skin areas
- ◆ True, Acrodermatitis enteropathica is due to Zinc deficiency (genetic or acquired)



Mental Retardation + Fair Complexion + ^{bad smell (or)} Mousy Odor of Urine

Phenylketonuria (PKU) - MC IEM worldwide.

↳ Inborn Error of Metabolism

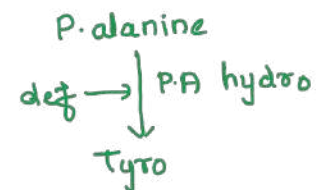
◆ Due to deficiency of: Phenylalanine hydroxylase

◆ Diagnosis

- Newborn screening (Guthrie test)
- ↑ Phenylalanine, ↓ Tyrosine in blood ← TMS ← IOL

◆ Treatment (Rx):

- Phenylalanine-restricted diet
- Tyrosine supplementation



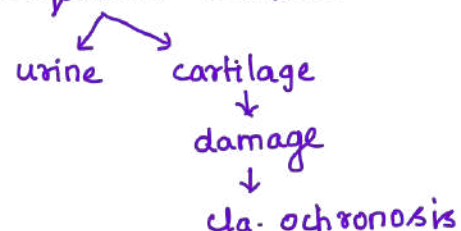
304 ☞ Joint Pain + Darkening of Ear Pinna + Urine Turns Dark on Standing

Alkaptonuria

- ◆ Due to deficiency of: Homogentisate oxidase
- ◆ Accumulated substance: Homogentisic acid → Benzoquinone acetate

◆ Treatment (Rx):

- Diet low in tyrosine and phenylalanine
- Vitamin C (may slow ochronosis)
- Symptomatic joint care; nitisinone is under study



BP
Abdominal Pain + Darkening of Urine on Sun Exposure } Acute intermittent porphyria
↓
def of PBG-deaminase.

Porphyria cutanea tarda (or another porphyria)

- ◆ Due to deficiency of: Uroporphyrinogen decarboxylase
- ◆ Treatment (Rx):
 - Avoid sunlight, phlebotomy, low iron diet, hydroxychloroquine
 - Discontinue triggers (alcohol, estrogen, iron)

Vomiting After Breastfeeding (7-Day-Old Infant)

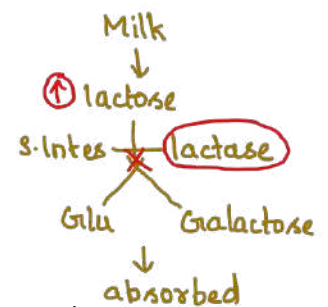
Galactosemia (classic)

- ◆ Due to deficiency of: Galactose-1-phosphate uridylyltransferase (GALT)
- ◆ Breastfeeding is Absolute C/I.

Abdominal Cramping and Diarrhea After Milk Intake (15-Year-Old)

Lactose intolerance

- ◆ Due to deficiency of: Lactase enzyme (brush border enzyme)
- ◆ Treatment (Rx):
 - Avoid milk products, lactose-containing foods, lactase enzyme supplements



Abdominal Cramp After Fruit Juice

Hereditary Fructose Intolerance

- ◆ Due to deficiency of: Aldolase B
- ◆ Diagnosis:
 - Positive urine fructose, hypoglycemia with fructose ingestion

Urine Fructose Positive but Asymptomatic

Essential Fructosuria

- ◆ Due to deficiency of: Fructokinase



Xanthelasma

→ def on LPL (or) APO C2
≡ Type ① → ↑ CM → ↑ TG → Pancreatitis
→ eruptive xanthoma.

Type II - Familial Hypercholesterolemia

$\uparrow$ LDL $\rightarrow$ Type IIa
 $\uparrow$ LDL + $\uparrow$ VLDL $\rightarrow$ Type IIb

Cholesterol: $\uparrow$ LDL cholesterol (normal triglycerides)

Mutation in: LDL receptor gene [Apo B100]

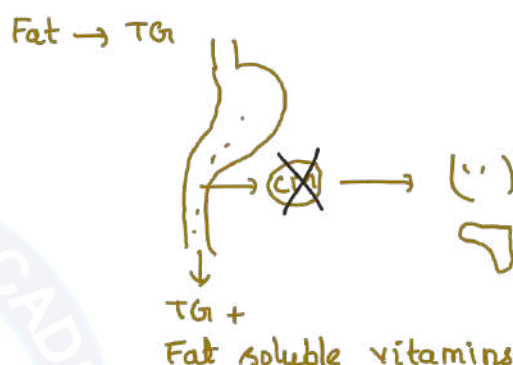
◆ **Inheritance:** Autosomal dominant

- Premature atherosclerosis
- Tendinous xanthomas (especially Achilles)
- Corneal arcus (under 50 y/o)

Abetalipoproteinemia $\Rightarrow$ $\downarrow$ CM, $\downarrow$ VLDL, $\downarrow$ LDL

Mutation in the MTTP gene (microsomal triglyceride transfer protein)

- ◆ No apoB-48 or apoB-100 synthesis
- ◆ No chylomicrons, VLDL, or LDL
- ◆ Steatorrhea, failure to thrive
- ◆ Acanthocytosis (spiky RBCs)
- ◆ Progressive ataxia and retinitis pigmentosa
- ◆ Treatment: Vitamin E supplementation



Pyruvate Dehydrogenase Complex (PDC)

Converts pyruvate $\xrightarrow{\text{mito}}$ acetyl-CoA

Requires 5 cofactors: TPP, Lipoic acid, CoA, FAD, NAD⁺ ("TLC For Nancy")

SUBUNIT	NAME	FUNCTION
E1 - B ₁	Pyruvate dehydrogenase	Decarboxylates pyruvate
E2 - B ₅ / lipoamide	Dihydrolipoyl transacetylase	Transfers acetyl group to CoA
E3 - B ₃ /B ₂	Dihydrolipoyl dehydrogenase	Regenerates oxidized lipoamide

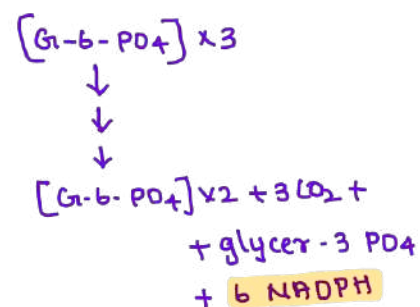
HMP Shunt (Pentose Phosphate Pathway) $\Rightarrow$ cytoplasm

RLS (Rate-Limiting Step): Glucose-6-phosphate dehydrogenase (G6PD)

◆ **NADPH** - used in:

- Fatty acid & cholesterol synthesis
- Glutathione reduction (RBCs)

◆ Ribose-5-phosphate - nucleotide synthesis



Hemolytic Anemia After ATT or Primaquine

◆ Primaquine or Antituberculosis drugs (dapsone, isoniazid), G6PD deficiency

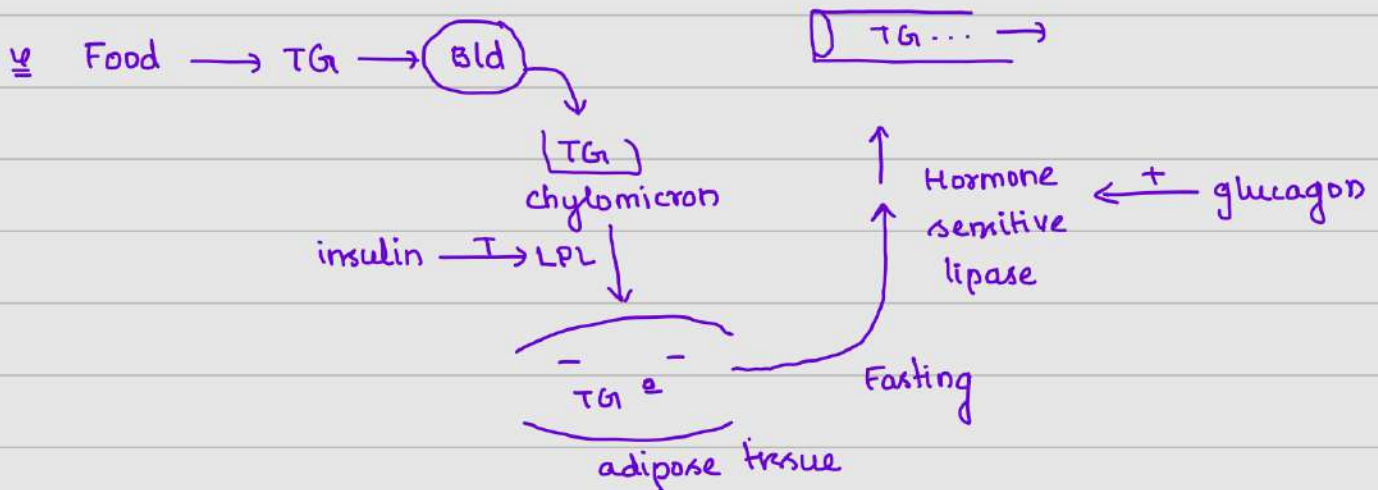
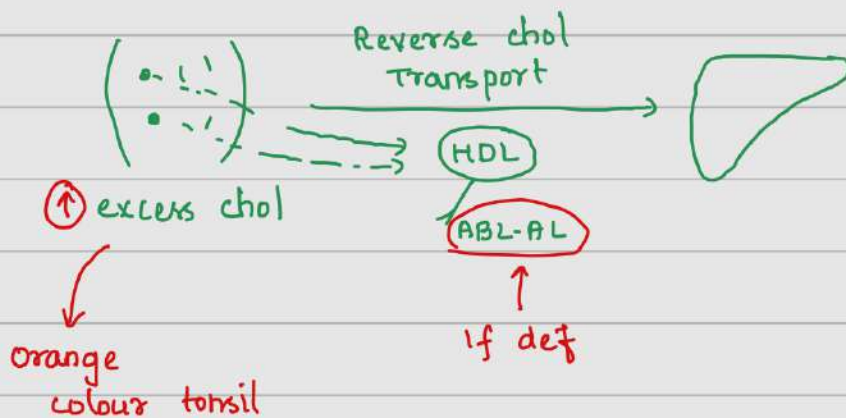
◆ **Mechanism:** $\downarrow$ NADPH $\rightarrow$ $\downarrow$ reduced glutathione $\rightarrow$ $\downarrow$ oxidative stress $\rightarrow$ RBC membrane damage and Leads to intravascular hemolysis.

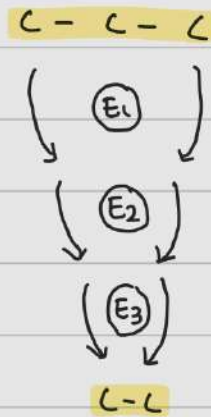
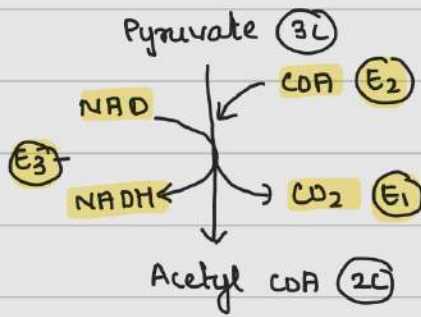
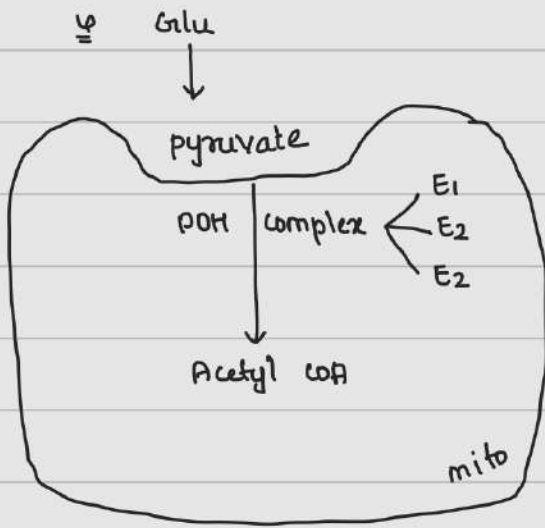
• x-linked recessive An*

anemia

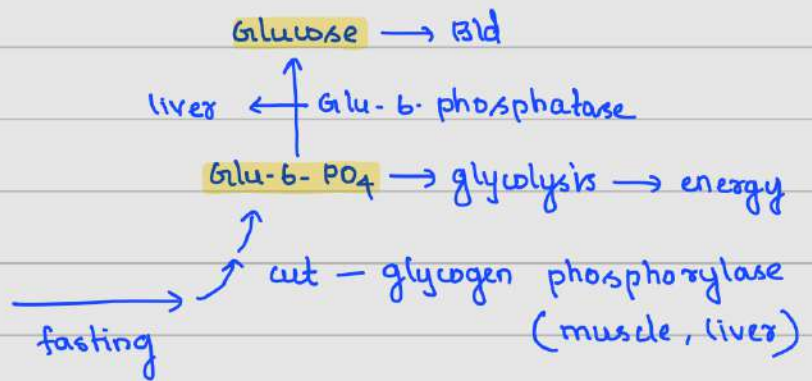
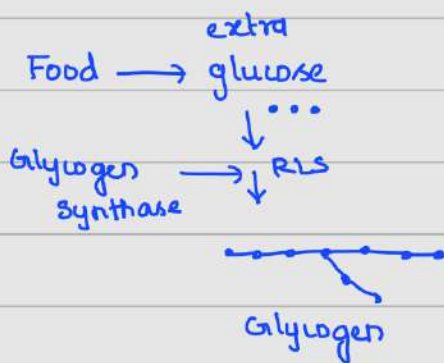
	defect	⬆ of	C/F
Type 1	Apo C2 / LPL	CM $\rightarrow$ TG $\uparrow$	Pancreatitis eruptive xanthomas
An* Type 2a 2b }	Apo B100 [LDL rec]	LDL $\uparrow$ LDL $\uparrow$ + VLDL $\uparrow$	Tendon xanthoma Atherosclerosis
Type 3	Apo E	$\uparrow$ remnant of cm / VLDL	-
Type 4	Over production of VLDL	TG $\uparrow$	-

≡ Tangier's disease:-

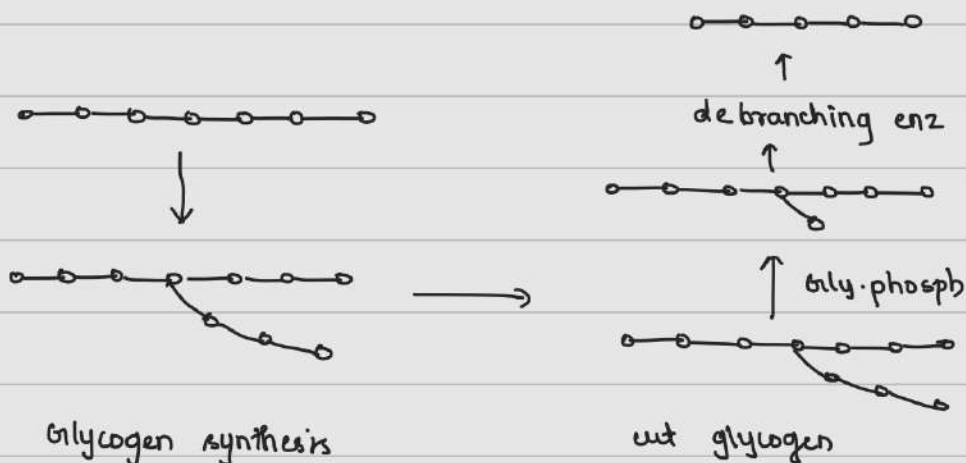




11/6



11/6



GLUCOSE TRANSPORTERS	
Glut 1	RANI - RBC , Blood brain barrier, Cornea, Kidney ⇒ <i>Widely distributed</i>
Glut 2	LAXMI - Liver, Pancreas Beta cells , kidney , Small intestine
Glut 3	BAI - Brain , Placenta
Glut 4	MADE - Muscle, adipocytes ⇐ <i>Insulin dependent</i>
Glut 5	INDIA - Intestine(Jejunum), Testes(Sperm), Microglia
SGLT 1	Small intestine (enterocytes), kidney (proximal tubule, S3 segment), heart
SGLT 2	Kidney (proximal tubule, S1/S2 segment)

Glycogen storage disorders

Type	Name	Enzyme def	Glycogen	Clinical
0		Glycogen synthase	None	Hypoglycemia, ketosis
CD AB	I * VonGierke	Glucose-6-phosphatase ↓ <i>only found in liver</i>	Normal	Severe hypoglycemia, big liver, Lactic acidosis
	II Pompe	Lysosomal α1, 4-glucosidase	Inclusion bodies with glycogen	Big heart, weakness, death
	III Cori	Debranching enzyme (α1,6-glucosidase)	Shorter outer branches	Mild hypoglycemia, big liver
	IV Anderson	Branching enzyme	Long chains with few branches	Cirrhosis, hypoglycemia, death
	V * McArdle	Muscle glycogen phosphorylase	Normal	Muscle cramps, weakness, teens
	VI Hers	Liver glycogen phosphorylase	Normal	Mild hypoglycemia, cirrhosis

- ◆ **Most Common Type (Mc type) Type I — Von Gierke disease**
- ◆ **Most Common Type in Adults Type V — McArdle disease**
- ◆ **Von Gierke Disease (Type I) due to Deficiency: Glucose-6-phosphatase**

≡ Von gierke → Hypoglycemia.

Pompe → acid maltase def ⇒ DCMP

Cori → debranching def ⇒ short branched glycogen

Ander → branching def ⇒ linear glycogen

McArdle → muscle cramp + dark urine

Von. Gierke disease :-

- Severe fasting **hypoglycemia** (child around 3 years old)
- Hepatomegaly
- **Doll-like facies**
- Hyperuricemia (due to increased nucleotide turnover)
- Hyperlipidemia

◆ **Features:** Hypoglycemia, lactic acidosis, hyperuricemia, and hepatomegaly

McArdle Disease (Type V) due to Muscle glycogen phosphorylase

- ◆ Exercise intolerance
- ◆ Muscle cramps and **myoglobinuria** after strenuous exercise
- ◆ No rise in lactate during exercise
- ◆ Normal blood glucose

GSD + ECG Defect

- ◆ **Pompe Disease** (Type II) Deficiency: **Lysosomal acid alpha-1,4-glucosidase (acid maltase)**
- ◆ Cardiomegaly, hypertrophic cardiomyopathy, muscle weakness, early death

Important points

◆ **Glycolysis**

- Conversion of **glucose to pyruvate**, **producing ATP and NADH**.
- Location: **Cytoplasm**

→ 7 ATP
anaerobic → 2 ATP
Rapoport leubing → ZERO

◆ **Anaerobic Glycolysis**

- Conversion of **glucose to lactate** when oxygen is low.
- Pyruvate → lactate via lactate dehydrogenase
- Regenerates NAD^+ for glycolysis to continue
- Yields **2 ATP** per glucose (net)
- Occurs in: RBCs, exercising muscle, hypoxic tissues

Glu
↓
Pyruvate
↓ lactate dehydrogenase
lactate

◆ **Peptide Bond Formation**

- Occurs during: Protein synthesis (translation)
- Enzyme: **Peptidyl transferase** (a ribozyme in the ribosome)

→ directly → ZERO
→ Indirectly → 4 P_0_4 required.

◆ **TCA Cycle (in Mitochondria matrix)**

- Input: Acetyl-CoA (from pyruvate via pyruvate dehydrogenase)
- Products per acetyl-CoA:

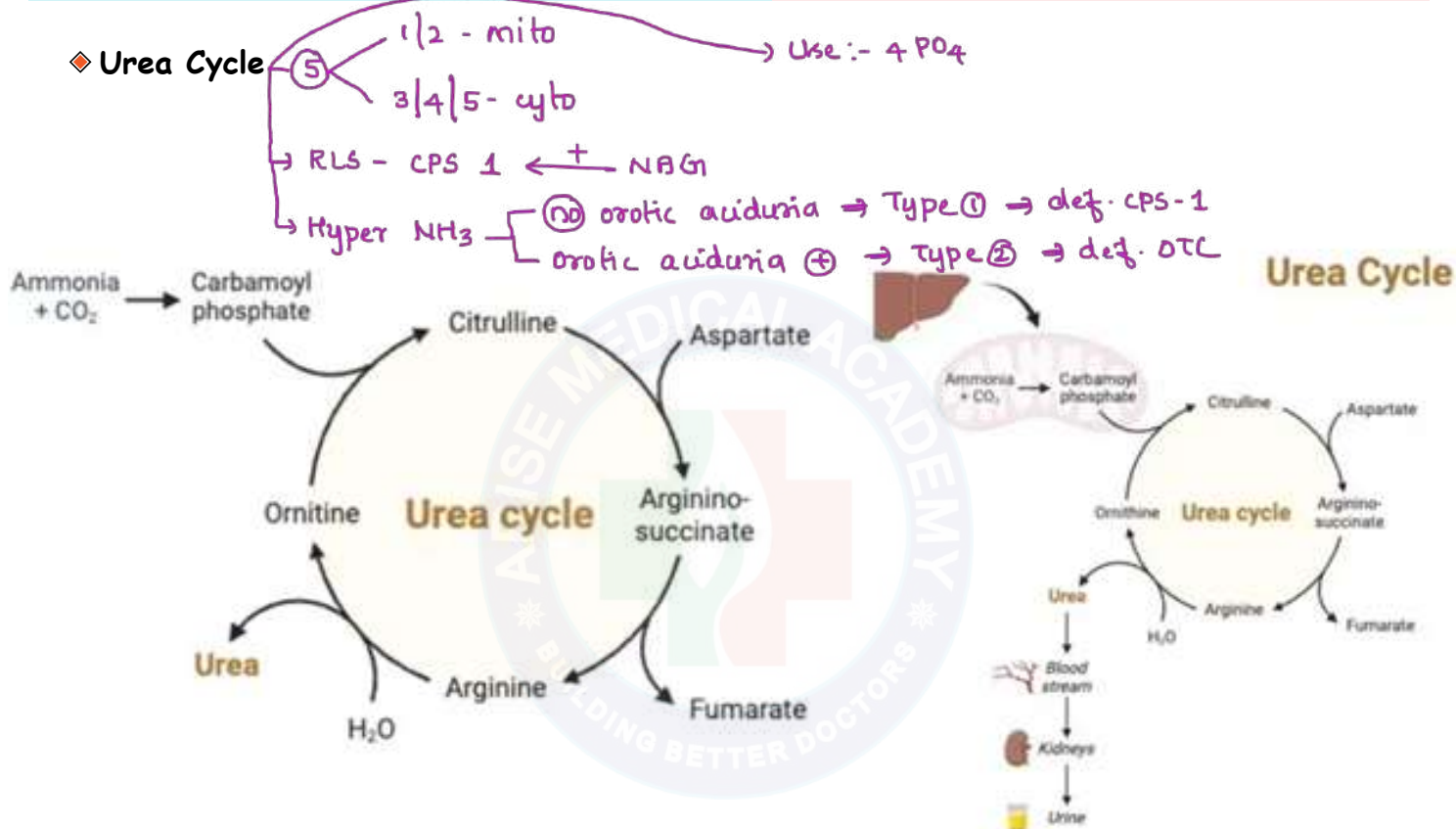
3 NADH 1 FADH₂ = 10 ATP
1 ATP 2 CO₂
↓ decarboxylation.

Acetyl CoA
↓
Oxaloacetate
→
3 NADH
1 FADH
1 ATP

♦ ATP Yield per 1 Molecule Glucose (Complete Oxidation) = 32 ATP

PROCESS	ATP YIELD (NET)
Glycolysis Glu → Pyruvate x2	7 ATP
Pyruvate ---> Acetyl-CoA	2 NADH = 5 ATP
TCA Cycle (2 turns)	10 ATP x2 = 20 ATP
Total	32 ATP

♦ Urea Cycle



- Detoxify ammonia by converting it to urea for excretion
- Location: Liver (mitochondria + cytosol)
- Key enzymes: Carbamoyl phosphate synthetase I (rate limiting), ornithine transcarbamylase. [OTC]

♦ Total 5 Enzymes of Urea Cycle

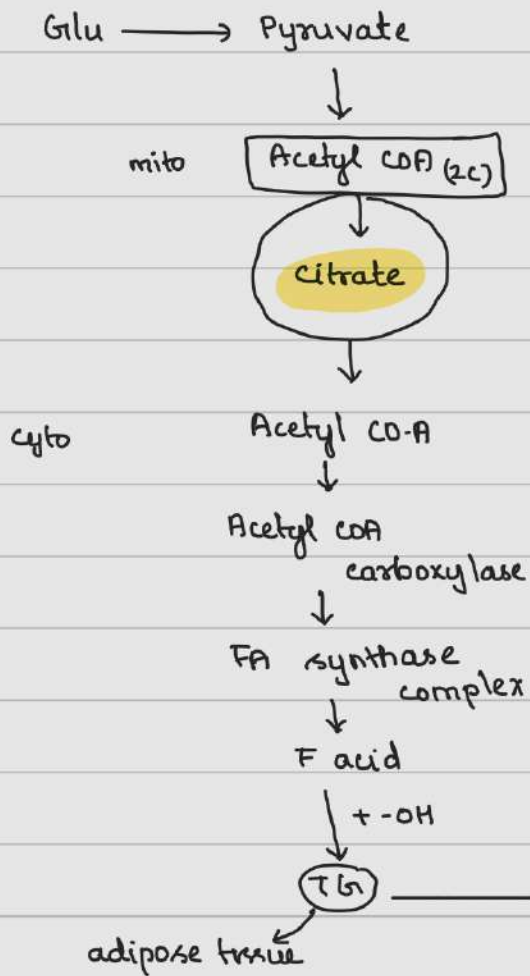
- Carbamoyl phosphate synthetase I (CPS I) → mito
- Ornithine transcarbamylase (OTC) → mito
- Arginosuccinate synthetase → cyto
- Arginosuccinate lyase → cyto
- Arginase → cyto

CPS II (Carbamoyl Phosphate Synthetase II)

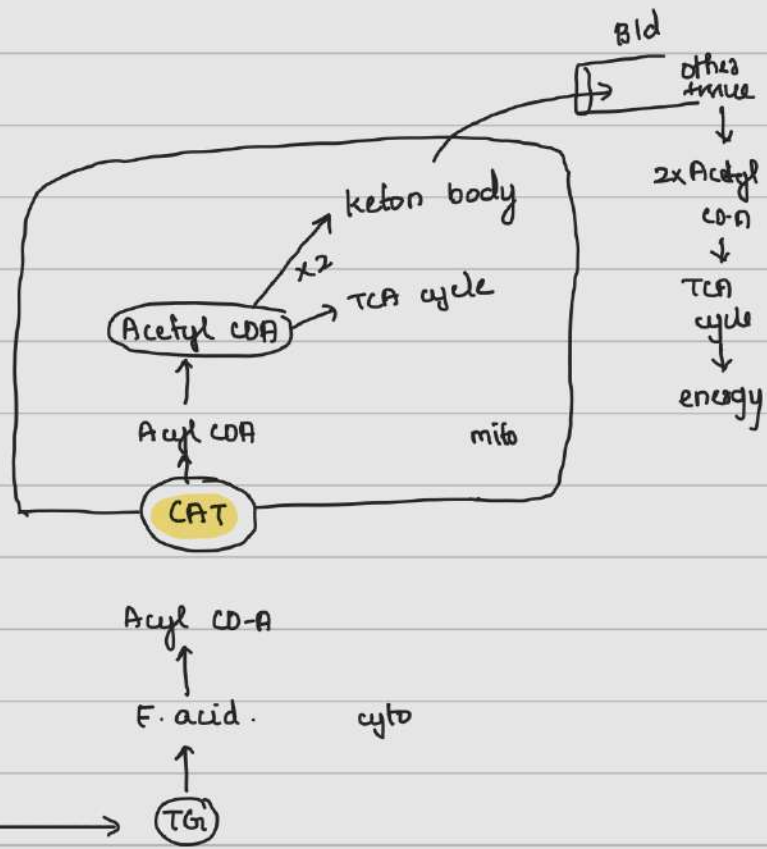
- ♦ Involved in pyrimidine synthesis, located in cytosol (not urea cycle)

(C|U|T)

WELL-Fed state



Fasting



Palmitic Acid CCCCCCCCCCCCCCCC(=O)O

- ◆ Type: Saturated fatty acid (16 carbons)
- ◆ Synthesis:
 - Occurs in cytoplasm
 - Uses acetyl-CoA, malonyl-CoA, NADPH
 - Enzyme: Fatty acid synthase complex

Hypoglycemia + ⊕ k. body
↓
defect in β-oxidation

FEATURE	HOMOCYSTEINURIA	MARFAN SYNDROME
Etiology	Inborn error of metabolism (cystathionine β-synthase deficiency) ^{an*}	Genetic disorder of connective tissue (FBN1 gene mutation) ^{ch. 15}
Inheritance	Autosomal recessive	Autosomal dominant
Lens Dislocation (Ectopia Lentis)	Downward displacement of the lens <i>Tall Arachnodactyl Flat sole</i>	Upward displacement of the lens
Intellectual Disability	Common	Rare
Thrombo-embolic Events	High risk ^{⊕⊕} (due to elevated homocysteine levels)	Uncommon [⊖]

Pancake Smell of Urine (Burnt Sugar Smell):

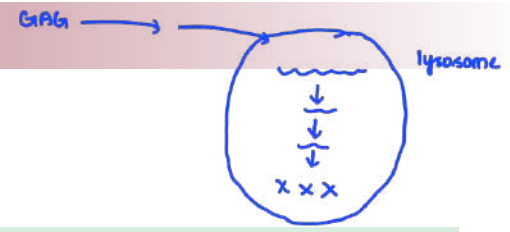
- ◆ Maple Syrup Urine Disease (MSUD)
- ◆ Deficiency of: Branched-chain alpha-ketoacid dehydrogenase complex
- ◆ Accumulated: Branched-chain amino acids (leucine, isoleucine, valine) and their ketoacids.

Renal Stones - Cystinuria

- ◆ Cause: Defect in renal tubular reabsorption of COLA amino acids:

- Cysteine
- Ornithine
- Lysine
- Arginine

 hexagonal crystals in urine



Hunter Disease vs Hurler Disease

Feature	Hunter Syndrome	Hurler Syndrome
		
Inheritance	X-linked recessive	Autosomal recessive
Enzyme Deficiency	Iduronate-2-sulfatase	Alpha-L-iduronidase
Accumulated GAGs	Heparan sulfate and dermatan sulfate	Heparan sulfate and dermatan sulfate
Clinical Features	Milder, no corneal clouding, aggressive behavior	Severe, corneal clouding, developmental delay

Tay-Sachs Disease

- Deficiency: Hexosaminidase A
- Accumulated substrate: GM2 ganglioside
- MCC of cherry-red spot on macula.

Cherry red spot + HSM → Sandhoff
 alt def Hexosaminidase B & B
 no HSM +



Pheochromocytoma

- Tumor of: chromaffin cell (of adrenal medulla)
 - Episodic refractory hypertension
 - Headache, sweating, palpitations (classic triad)
- Definitive treatment: Surgical removal after alpha-blockade (phenoxybenzamine) followed by beta blockers.

Lesch-Nyhan Syndrome

- Enzyme Defect: HGPRTase
 - Clinical Features:
 - Orange sand-like crystals in urine due to uric acid
 - Gout and kidney stones
 - Self-mutilation behaviors (e.g., lip and finger biting)
- MR, x-link recessive

(HGPRT)



Methylated (or) NE

- ## Nutrition and Biochemistry

RDA dietary fiber	~25-30 g/day (varies by age/gender) → 40g/day
GAG in synovial fluid	Hyaluronic acid
GAG with anticoagulant property	Heparan SO ₄
GAG without uronic acid	→ keratan SO ₄
GAG helping corneal transparency	→

◆ **Amino acid** (protein) metabolism, Specifically, the **nitrogen** from amino acid deamination is converted to **urea** in the **liver** via the **urea cycle**.

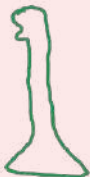
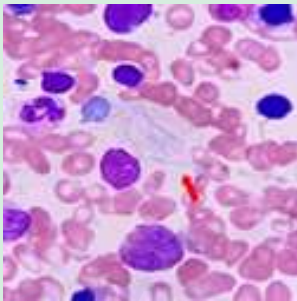
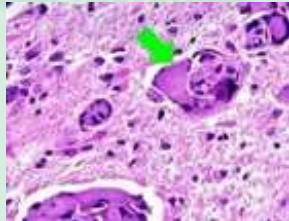
- ◆ Melatonin is Formed by Tryptophan $\rightarrow$ Serotonin $\rightarrow$ Melatonin, Produced mainly in the pineal gland $\rightarrow$ *Regulate sleep cycle*
- ◆ Tryptophan is an Important Growth Factor for Niacin (Vitamin B3) synthesis, Also a precursor for serotonin and melatonin $\rightarrow$ *Growth of salmonella*
- ◆ The lac operon encodes proteins that allow E. coli to use lactose and allolactose (an inducer)
 - Allolactose binds to the repressor and removes it, allowing transcription

E. coli → Glucose →
 ↘ lactose → enz

- ### ◆ Composition of Creatine/ Creatinine

- ▶ **G** - Glycine
- A** - Arginine
- M** - Methionine.

- ◆ Most Common Amino Acid in Collagen Glycine (every third residue)
 - ▶ Collagen has a unique **Gly-X-Y** repeating structure (X and Y often **proline** or **lysine**)
- ◆ **Bond Resistant to Denaturation:** **peptide bond**.
- ◆ Turns and Bends Are Made Up Of **Proline and glycine** residues (commonly found in beta-turns)
- ◆ Cyanide Poisoning Treatment: Hydroxycobalamin or sodium thiosulfate / **NaHCO_3**
- ◆ Bitter Almond Smell of Breath - Cyanide poisoning
 - ▶ Cyanide inhibits Complex IV (cytochrome c oxidase) in ETC

DISEASE	ENZYME DEFICIENCY	ACCUMULATED SUBSTRATE	CLINICAL FEATURES
Tay-Sachs Disease	Hexosaminidase A	GM2 ganglioside 	Most common to cause Cherry-red spot on macula, developmental delay, seizures, blindness, death by age 3-5. No hepatosplenomegaly.
Gaucher Disease	Glucocerebrosidase 	Glucocerebroside 	Most common LSD, Hepatosplenomegaly, bone pain, "Erlenmeyer flask" deformity, pancytopenia. Lipid laden macrophages crumpled tissue paper appearance.
Fabry Disease X-rec	Alpha-galactosidase A	Globotriaosylceramide 	Angiokeratomas, corneal dystrophy, acroparesthesias, renal and cardiac involvement.
Krabbe Disease X-rec	Galactocerebrosidase	Galactocerebroside 	Peripheral neuropathy, Developmental regression, irritability, spasticity, CN II, optic atrophy. Globoid cells
Niemann-Pick Disease	Sphingomyelinase	Sphingomyelin	Hepatosplenomegaly, cherry-red spot, neurodegeneration, foam cells in bone marrow. • zebra body

Gaucher :- Pancytopenia + HSM.

⊗-Fabry :- angiokeratoma

⊗-Hunter :- no corneal clouding

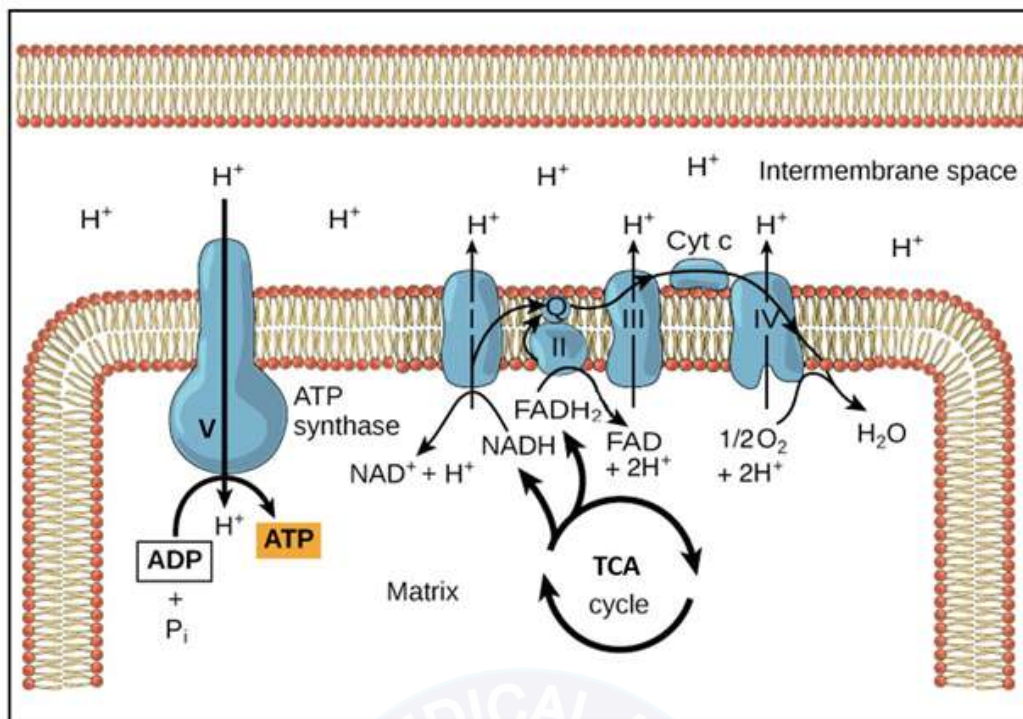
Hurler :- corneal clouding

Neiman pick :- Cherry red + HSM + foam cell | zebra

Tay Sachs :- Cherry red + no HSM

Sandhoff :- Cherry red + HSM

ETC (Electron Transport Chain) Location



◆ Present in the inner mitochondrial membrane

40 42

COMPLEXES OF ETC	INHIBITORS
4H ⁺ Complex I(NADH dehydrogenase)	Rotenone, amobarbitol, biguanide, Piercidin A (Fish poison)
Zero Complex II(Succinate dehydrogenase)	An* Carboxin, Malonate, TTFA
4H ⁺ Complex III(Cytochrome bc1 complex)	Antimycin A, BAL
2H ⁺ Complex IV (Cytochrome c oxidase)	CO, cyanide, H ₂ S, Sodium azide
Complex V [ATP synthase]	Atractyloside, Oligomycin, Venturicidin

ETC Uncouplers

► 2,4-Dinitrophenol (DNP), aspirin (high doses), thermogenin (in brown fat)

◆ Bullet retained in body can lead to poisoning of Lead poisoning (Pb) — lead can be retained in bones and tissues causing toxicity

◆ cAMP is formed from ATP by the enzyme adenylyl cyclase ATP $\xrightarrow{\text{A-cyclase}}$ cAMP

◆ Gluconeogenesis is mainly seen in Liver → Kidney cortex, Not significant in muscle

◆ Branched-chain amino acids are Leucine, Isoleucine, and Valine. [LIV]

∴ Not a substrate of gluconeogenesis ⇒ Acetyl CoA

◆ Cofactor for xanthine oxidase: **Molybdenum**.

◆ Inhibitors of xanthine oxidase: Allopurinol
(used to treat gout by decreasing uric acid production)

◆ Cofactor for Pyruvate Kinase and Phosphofructokinase-1 (PFK-1): Mg^{2+} (Magnesium ion)

◆ Coenzyme for Transaminase: Pyridoxal phosphate (Vitamin B6)

→ decarboxylation → Vit B6
→ Glycogen phosphorylase → Vit B6

◆ Isozymes are

► Enzymes with the same function but different structure
(different amino acid sequences or subunit composition)

NH_3 → transporter from muscle → alanine

◆ Ammonia Detoxified/Transported from Brain in the form of Glutamine.

◆ Essential Fatty Acids: Linoleic acid (omega-6), Alpha-linolenic acid (omega-3)

◆ Apoprotein for Chylomicron Remnant - Apo E

Apo in cm { Apo B48
Apo C2
Apo E

◆ ApoE4 is Associated with Alzheimer's disease (increased risk)

◆ Assembly of Chylomicrons Occurs in the intestinal mucosal cells (enterocytes)

◆ Cardioprotective Fatty Acid - Omega-3 fatty acids (EPA and DHA)

◆ Keshan Disease is Caused by Deficiency of selenium

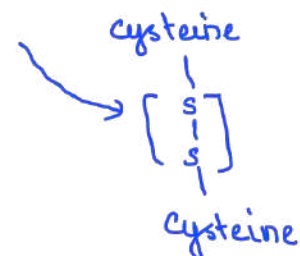
◆ Start Codon is AUG (codes for methionine)

◆ Vitamin K is Required for Gamma-carboxylation of glutamic acid residues in clotting factors II, VII, IX, X and proteins C and S

◆ Cardio Protective Cholesterol is HDL cholesterol

◆ Disulfide Bonds Formed Between Cysteine residues in proteins

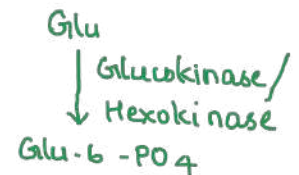
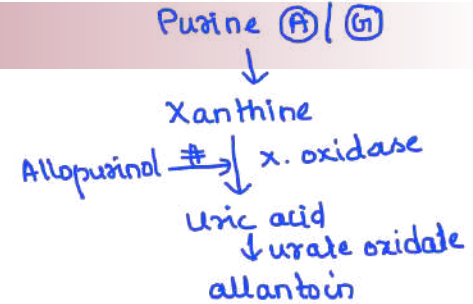
◆ Sulfur-Containing Amino Acids Cysteine and Methionine



★ Steps of PCR (Polymerase Chain Reaction)

- Denaturation (94-98°C) — DNA strands separate
- Annealing (50-65°C) — primers bind to target sequences
- Extension (72°C) — Taq polymerase synthesizes new DNA strands

◆ Enolase is Inhibited by NaF (F^-) (used in glycolysis inhibition in lab tests)



- ◆ Prostaglandins are Obtained From **Arachidonic acid** (a polyunsaturated fatty acid).
- ◆ **Antibody Working as Enzyme** Called as **Abzyme** (antibody with catalytic activity).
- ◆ Function of **Arginine** - Precursor of **Nitric oxide (NO)**, which causes vasodilation
- ◆ **Winter Chest Pain**; Defect it is Likely Variant (**Prinzmetal**) **Angina** due to coronary artery vasospasm
 - Associated with endothelial dysfunction and impaired NO production
- ◆ **Menke's Disease** is Due To Defect in **ATP 7A** (copper transporting ATPase) ---> leads to copper deficiency 29. is Associated With Wilson's disease (copper accumulation disorder)
- ◆ **Zellweger Syndrome is Associated With** **Peroxisomal enzy**
 - Peroxisomal biogenesis defect --> accumulation of very long chain fatty acids an*
- ◆ **Vitamin A Storage**
 - Stored mainly in hepatic stellate (Ito) cells of the liver
- ◆ N-Base Present in RNA but Absent in DNA - **U** [**uracil**]
- ◆ DNA Double Helix is Held By Hydrogen bonds between complementary base pairs
- ◆ **Purine --> Hypoxanthine --> Xanthine --> Uric acid --> allantoin**
- ◆ Purine --> Uric Acid Converted to Allantoin by **Uricase (urate oxidase)** (absent in humans)
- ◆ Rate-Limiting Enzyme in Cholesterol Synthesis **HMG-CoA reductase**
- ◆ Atorvastatin Inhibits Their Target Enzyme By Competitive inhibition of **HMG-CoA reductase**
- ◆ Final Product of Purine Metabolism - **uric acid**.
- ◆ Limiting Amino Acids in Wheat - **lumin**.
- ◆ **Vitamin B12 Status** Measured By **Serum methylmalonic acid** and **homocysteine levels** (both increase in deficiency)
- ◆ **Nucleotide Excision Repair Defect** - **Xeroderma pigmentosa** pigmentation. Facial cancer skin.
- ◆ **Mismatch Repair Defect** - **HNPCC [Lynch syn]**
- ◆ Fatty Acid Transport Through Mitochondrial Membrane Is Via **CAT** shuttle.
- ◆ DNA Packaging Is Done By **histone**.
- ◆ A **Paint Factory Worker** Complaining of Joint Pain, Insomnia, Fatigue, Constipation, Headache, and Anemia
 - Lead poisoning
 - Enzyme affected: Aminolevulinic acid dehydratase (ALAD) and Ferrochelatase (in heme synthesis)
- ◆ **Rash + History of Raw Egg Consumption**
 - Suggestive of Biotin deficiency (raw egg whites contain avidin which binds biotin)
 - Clinical features: dermatitis, alopecia, conjunctivitis